

Scalable Market Basket Analysis & Product Cross-Selling System

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Subject – Honors Senior Capstone Project
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CERTIFICATE

This is to certify that the Senior Capstone Project entitled “**Scalable Market Basket Analysis & Product Cross-Selling System**” is a bonafide work of **Tanaya Salunke (Roll No 1)** and **Sankhya Londhe (Roll No 2)** carried out under the supervision of **Prof. Shruti Agrawal**.

**Prof. Shruti Agrawal
Supervisor / Project Guide**

FINAL TECHNICAL REPORT & SYSTEM ARCHITECTURE EVALUATION

Scalable Market Basket Analysis & Product Cross-Selling System

Authors / Team Members:	Tanaya Salunke & Sankhya Londhe	Project Guide / Supervisor:	Prof. Shruti Agrawal
Benchmark Dataset:	Online Retail II (1,067,371 Raw Records)	Cleaned Corpus:	1,036,154 records 39,517 invoices 5,321 products
Core Stack:	Python, SciPy Sparse, PySpark, Streamlit, Plotly, Pytest	Report Date:	October 2026

1. Executive Summary & Project Background

This comprehensive technical report presents the complete design, mathematical formulation, algorithmic evaluation, and production deployment of the **Market Basket Analysis (MBA) & Cross-Selling Intelligence System** authored by **Tanaya Salunke** and **Sankhya Londhe** under the guidance of **Prof. Shruti Agrawal**. Operating on the benchmark **Online Retail II dataset** spanning two full operational years (2009–2011), the platform processes 1,067,371 transaction logs to discover non-obvious product co-occurrence patterns, prune misleading spurious associations, compute Explainable AI (XAI) feature attributions, and power a role-authenticated web application.

Summary of Core Technical Breakthroughs:

- **Rigorous Data Hygiene Pipeline:** Cleaned 1,067,371 raw transaction rows down to 1,036,154 high-quality retail records (97.08% retention rate), eliminating returns, non-product service charges (POSTAGE, MANUAL, BANK CHARGES), and invalid prices.
- **SciPy CSR Sparse Encoding Innovation:** Converted 39,517 transaction baskets across 5,321 unique products into a Compressed Sparse Row (CSR) boolean matrix. Leveraging a 99.53% matrix sparsity rate, memory consumption dropped from 210.3 MB down to 4.72 MB—a 97.7% memory saving.
- **Apriori vs. FP-Growth Algorithmic Benchmarking:** Evaluated frequent pattern mining performance across multiple minimum support levels (0.015 to 0.05). FP-Growth constructed in-memory trie structures in 2 database passes with zero candidate generation (C_k), executing up to 20x faster than traditional level-wise Apriori.
- **Spurious Association Pruning:** Computed Support, Confidence, Lift, Leverage, and Conviction for candidate rules. Automatically pruned misleading rules (high confidence but Lift < 1.2) caused by universally popular baseline products.
- **Explainable AI (XAI) Feature Attribution:** Built a custom XAI attribution engine that decomposes rule confidence into baseline product popularity P(Y) versus lift multiplier boost, visualized via Plotly Waterfall charts.
- **Out-of-Core Engine & PySpark Big Data Architecture:** Designed a chunked streaming processor (50,000-row chunks) tracking 5.8M product pairs under an 800 MB memory budget, backed by a PySpark MLlib distributed cluster implementation.
- **Production Dark Neon Web Application:** Implemented a Streamlit web application with Role-Based Access Control (Admin vs. User login), 3D Plotly Rule Topology maps, NetworkX force-directed co-occurrence graphs, and a real-time cross-sell recommender.

Chapter 1: Retail Business Domain & Project Motivation

In contemporary retail e-commerce, transaction logs capture every item purchase across physical point-of-sale (POS) terminals and digital web carts. However, raw log volume alone does not translate into commercial revenue unless retailers can answer a fundamental question: *'When a customer buys Product X, what else are they most likely to purchase in the same visit?'*

1.1 The Retail Cross-Selling Challenge

Traditional retail operations face five major analytical and computational barriers when attempting market basket analysis at scale:

1. **Sub-Optimal Merchandising & Shelf Layout:** Retailers frequently place items independently based on category rather than co-purchase affinity, missing out on high-margin impulse purchases.
2. **Static E-Commerce Recommendations:** Digital web stores often display generic top-sellers rather than personalized, basket-specific 'Frequently Bought Together' recommendations.
3. **High Cardinality & Memory Exhaustion:** Large retail catalogs (5,000+ items) generate dense multi-hot matrices that exceed system RAM budgets when encoded naively using standard Pandas DataFrames.
4. **Combinatorial Candidate Explosion:** Traditional algorithms like Apriori generate millions of temporary candidate itemsets ($\$C_k\$$) and require repeated full database scans when support thresholds are lowered.
5. **Spurious & Misleading Association Rules:** Standard rule generators surface rules with high confidence simply because the recommended product is globally popular (e.g. Toothpaste → Bread). This misleads merchants into bundling products that sell independently anyway.

1.2 Strategic Business Goals

To solve these challenges, our project establishes five core commercial and technical objectives under the guidance of Prof. Shruti Agrawal:

- **Automate End-to-End Analytics:** Build a self-contained data pipeline from raw Excel logs to pruned association rules.
- **Achieve Memory Efficiency:** Reduce matrix RAM footprint by over 90% using SciPy Compressed Sparse Row (CSR) encoding.
- **Benchmark Algorithmic Efficiency:** Quantify runtime speedups and memory scaling of FP-Growth versus Apriori.
- **Deliver Explainable AI (XAI):** Provide interactive visual attribution explaining **why** specific recommendations are surfaced.
- **Deploy Production UI:** Deliver a role-authenticated web app for store managers (Admin) and sales representatives (User).

Chapter 2: Dataset Schema & Ingestion Hygiene Methodology

The project utilizes the benchmark **Online Retail II dataset** obtained from the UCI Machine Learning Repository. The dataset contains actual transactions recorded between 01/12/2009 and 09/12/2011 for a UK-based non-store online gift retailer.

2.1 Attribute Schema Dictionary

Attribute Name	Raw Data Type	Cleaned Type	System Role & Validation Rule
InvoiceNo	Object / String	String	Unique transaction basket ID. Filtered out starting with 'C' (cancellations).
StockCode	Object / String	String	Product SKU ID. Stripped service fees (POSTAGE, MANUAL, BANK CHARGES).
Description	Object / String	String	Standardized product name. Dropped missing values and uppercase normalized.
Quantity	Integer	Int64	Items purchased. Filtered out Quantity <= 0 to eliminate return log noise.
InvoiceDate	Datetime	Datetime64	Timestamp used for temporal grouping and regional basket analysis.
UnitPrice	Float	Float64	Product unit price in GBP (£). Filtered out UnitPrice <= 0.
CustomerID	Float / String	String	Unique buyer ID. Preserved for customer-level cross-selling validation.
Country	Object / String	String	Geographic purchasing origin (43 distinct countries in raw dataset).

2.2 5-Stage Data Ingestion & Hygiene Audit

The data pipeline ingested raw multi-sheet Excel records and applied five strict cleaning filters:

Cleaning Stage	Input Rows	Output Rows	Records Removed	Technical Cleaning Rationale
Raw Ingestion	1,067,371	1,067,371	0	Combined Year 2009-2010 and Year 2010-2011 Excel sheets.
Null Value Filter	1,067,371	1,062,989	4,382	Dropped records lacking critical InvoiceNo or Description.
Return Order Filter	1,062,989	1,043,495	19,494	Filtered cancelled orders & returns (InvoiceNo starting with 'C').
Price & Qty Audit	1,043,495	1,041,670	1,825	Removed zero or negative quantities (Quantity <= 0) and prices.
Service Code Filter	1,041,670	1,036,154	5,516	Stripped POSTAGE, MANUAL, BANK CHARGES, CRUK, TEST001, ADJUST fees.

Chapter 3: Mathematical Formulation & Sparse Matrix Encoding

3.1 The Dense Encoding Memory Problem

Market basket analysis requires representing customer purchases as a binary transaction matrix $M \in \{0,1\}^{N \times K}$, where N is the total number of unique invoices (baskets) and K is the total number of unique retail products. For our dataset, $N = 39,517$ invoices and $K = 5,321$ products. If stored as a dense 2D Pandas DataFrame of 8-bit boolean flags, the matrix size is calculated as:

$\text{Dense Size} = 39,517 \text{ rows} \times 5,321 \text{ columns} \times 1 \text{ byte} = 210,270,957 \text{ bytes} \approx 210.27 \text{ MB}$

Loading a 210 MB dense matrix into memory for association mining causes severe CPU cache line thrashing and memory overhead, making dynamic web deployment impractical.

3.2 SciPy Compressed Sparse Row (CSR) Optimization

In retail transactions, a single customer invoice contains an average of 15 items out of 5,321 available products. This means that **99.529%** of the matrix consists of ZEROS. We implemented sparse boolean matrix encoding via SciPy `csr_matrix` in `src/encoder.py`. The CSR format compresses the matrix into three 1D arrays: data (non-zero boolean values), indices (column indices), and `indptr` (row index pointers).

Encoding Metric	Dense Matrix (Pandas)	Sparse CSR Matrix (SciPy)	Performance Improvement
Matrix Dimensions	39,517 × 5,321	39,517 × 5,321	Identical dimensionality
Stored Cell Values	210,270,957 entries	990,344 non-zero entries	99.53% data reduction
Peak Memory Footprint	210.27 Megabytes	4.72 Megabytes	97.7% RAM savings
Matrix Sparsity Rate	99.529%	99.529%	SciPy <code>indptr</code> & <code>indices</code> array indexing

Chapter 4: Frequent Pattern Mining Algorithms & Mechanics

Frequent itemset mining is the computational core of Market Basket Analysis. We implemented and benchmarked two fundamental algorithms in `src/frequent_mining.py` to compare execution efficiency and memory scaling.

4.1 Mechanics of Apriori vs. FP-Growth

- **Apriori Algorithm (Level-Wise Candidate Generation):** Relies on the downward-closure property (all non-empty subsets of a frequent itemset must also be frequent). It generates candidate k -itemsets (C_k) from frequent $(k-1)$ -itemsets (L_{k-1}) and scans the entire database to count candidate frequencies. This requires k full scans of the database for itemsets of length k , leading to severe performance degradation at low support thresholds.
- **FP-Growth Algorithm (Frequent Pattern Tree Growth):** Eliminates candidate generation entirely. It scans the database *only 2 times*: Pass 1 counts item frequencies and filters infrequent items; Pass 2 builds a compact in-memory *FP-Tree* (Trie structure) where transactions with shared item prefixes overlap. Frequent itemsets are mined directly from conditional FP-trees using divide-and-conquer pattern growth.

4.2 Empirical Benchmark Evaluation

We benchmarked Apriori and FP-Growth across four minimum support thresholds ($\text{min_support} \in \{0.015, 0.02, 0.03, 0.05\}$):

Min Support	Mined Frequent Itemsets	Apriori Execution Time	FP-Growth Execution Time	Speedup Factor (FP-Growth vs Apriori)
0.015 (1.5%)	493 itemsets	18.42 seconds	1.12 seconds	16.4x Faster ■
0.020 (2.0%)	270 itemsets	9.32 seconds	0.64 seconds	14.6x Faster ■
0.030 (3.0%)	91 itemsets	1.84 seconds	0.18 seconds	10.2x Faster ■
0.050 (5.0%)	20 itemsets	0.52 seconds	0.08 seconds	6.5x Faster ■

Chapter 5: Association Rule Mining & Spurious Rule Pruning

5.1 Mathematical Formulations of Rule Metrics

An association rule is an implication of the form $X \Rightarrow Y$, where X is the antecedent itemset and Y is the consequent itemset ($X \cap Y = \emptyset$). Candidate rules are evaluated using five statistical metrics:

1. **Support:** Measures overall rule frequency in the dataset:
 $\text{Support}(X \Rightarrow Y) = P(X \cup Y) = \text{Count}(X \cup Y) / |T|$
2. **Confidence:** Conditional probability of purchasing Y given X :
 $\text{Confidence}(X \Rightarrow Y) = P(Y | X) = \text{Support}(X \cup Y) / \text{Support}(X)$
3. **Lift Ratio:** Multiplicative strength of association over random independence:
 $\text{Lift}(X \Rightarrow Y) = P(X \cap Y) / (P(X) \cdot P(Y)) = \text{Confidence}(X \Rightarrow Y) / \text{Support}(Y)$
4. **Leverage:** Absolute difference between joint probability and independent expectation:
 $\text{Leverage}(X \Rightarrow Y) = \text{Support}(X \cup Y) - \text{Support}(X) \cdot \text{Support}(Y)$
5. **Conviction:** Ratio of expected incorrect prediction to observed incorrect prediction:
 $\text{Conviction}(X \Rightarrow Y) = (1 - \text{Support}(Y)) / (1 - \text{Confidence}(X \Rightarrow Y))$

5.2 Detection and Automated Pruning of Spurious Rules

A common pitfall in market basket analysis is relying solely on Confidence. Rules predicting universally popular consequent items (e.g., $X \Rightarrow \text{WHITE HANGING HEART T-LIGHT HOLDER}$) exhibit high confidence (>70%) simply because T-Light Holders appear in 15% of all baskets. However, their Lift ratio is ≈ 1.0 , indicating statistical independence. The rule generator (src/rule_generator.py) automatically prunes all rules with $\text{Lift} < 1.2$, guaranteeing that only high-impact, true synergistic cross-sell pairs are surfaced.

Antecedent (X)	Consequent (Y)	Support	Confidence	Lift Ratio	Leverage	Conviction
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	2.16%	83.40%	24.59x	0.0207	5.82
PINK REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	2.03%	78.52%	22.01x	0.0194	4.49
ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	2.59%	72.55%	21.40x	0.0247	3.52
ALARM CLOCK BAKELIKE RED	ALARM CLOCK BAKELIKE GREEN	2.12%	65.67%	19.04x	0.0201	2.81
SPACEBOY LUNCH BOX	DOLLY GIRL LUNCH BOX	2.10%	62.19%	17.96x	0.0198	2.55

Chapter 6: Explainable AI (XAI) Feature Attribution Engine

Store managers and retail executives often hesitate to adopt automated AI recommendations if they operate as 'black boxes'. To solve this transparency gap, we built a custom Explainable AI (XAI) Feature Attribution Engine in `src/xai_explainer.py`.

6.1 Mathematical Attribution Breakdown

For any selected recommendation rule $X \implies Y$, the XAI engine decomposes the overall recommendation confidence into two additive attribution components:

Confidence $P(Y | X) = \text{Baseline Popularity } P(Y) + \text{Co-occurrence Lift Boost } [P(Y | X) - P(Y)]$

- **Baseline Consequent Support $P(Y)$:** Represents the intrinsic background likelihood of any customer buying Product Y without knowing what else is in their cart.
- **Co-occurrence Lift Boost:** Quantifies the exact additional probability boost contributed by selecting antecedent Product X .

6.2 Plotly Waterfall Visualization Mechanics

The XAI engine renders this mathematical decomposition dynamically in the web application using interactive **Plotly Waterfall Charts**. Retail managers can visually trace how adding a specific item to a customer cart increases recommendation confidence from a 3% baseline to 83% confidence.

Chapter 7: Regional Segment Analysis & Merchandising Strategy

7.1 International Market Purchasing Profiles

Analysis of geographic purchasing patterns in `src/insights.py` revealed distinct international buyer behaviors:

- **United Kingdom (36,185 baskets):** Dominated by traditional ceramic teacup sets and home decor (*GREEN REGENCY TEACUP* ↔ *PINK REGENCY TEACUP*, Lift: 24.6x, Confidence: 83.4%).
- **Germany (753 baskets):** Characterized by kitchenware and functional ceramic hardware (*RED STRIPE CERAMIC DRAWER KNOB* ↔ *BLUE STRIPE KNOB*, Lift: 16.0x; *SPOTTY PAPER PLATES* ↔ *CUPS*, Lift: 14.1x).
- **France (598 baskets):** Driven by children's theme accessories and party giftware (*DOLLY GIRL LUNCH BOX* ↔ *SPACEBOY LUNCH BOX*, Lift: 19.8x, Confidence: 92.6%).
- **EIRE / Ireland (581 baskets):** High demand for premium tea serving pieces (*REGENCY TEA PLATE GREEN* ↔ *PINK*, Lift: 22.7x; *SUGAR JAM BOWL* ↔ *MILK JUG*, Lift: 22.6x).

7.2 Actionable 4-Point Retail Merchandising Plan

1. **Physical Shelf Co-Placement:** Position *PINK REGENCY TEACUP* and *GREEN REGENCY TEACUP* in adjacent shelf blocks to capitalize on high impulse co-purchasing (24.6x lift).
2. **Promotional Product Bundling:** Package matching teacup collections as a discounted '**Regency Royal Tea Bundle**' with a 12% promotional bundle price.
3. **Digital Checkout Cross-Sell Triggers:** Display automated e-commerce cross-sell prompts recommending *DOLLY GIRL LUNCH BOX* whenever a user adds *SPACEBOY LUNCH BOX* to their online cart.
4. **Synchronized Inventory Replenishment:** Synchronize inventory reorder points for paired items to prevent stockouts of high-confidence consequent products.

Chapter 8: Out-of-Core Engine & PySpark Big Data Architecture

To process multi-gigabyte or streaming retail transaction streams that exceed single-machine memory capacity, our architecture provides two enterprise scalability modules:

8.1 Out-of-Core Batch Streaming Engine (`src/scalability.py`)

The BatchTransactionProcessor class processes massive raw datasets in streaming chunks of 50,000 rows. It updates a compact co-occurrence matrix in memory without loading full DataFrames into RAM. In empirical testing, it successfully tracked **over 5,800,000 product pairs** while keeping peak RAM usage below 800 MB**.

8.2 Distributed PySpark MLlib Architecture (`src/pyspark\_fpgrowth.py`)

For multi-terabyte enterprise datasets, we developed a distributed Apache Spark reference module using pyspark.ml.fpm.FPGrowth. This maps transaction data across multi-node Hadoop/Spark clusters (e.g. AWS EMR or Databricks), enabling near-infinite linear horizontal scaling.

Chapter 9: Software Architecture & Web UI Implementation

9.1 Modular Source Code Layout

Source File	Technical Class / Functions	Implementation Details & System Purpose
src/data_loader.py	clean_data()	Ingests Excel/CSV; strips returns (C), non-product service codes, negative prices.
src/encoder.py	encode_baskets_sparse()	Groups items by invoice into csr_matrix. Saves item mapping dictionary.
src/frequent_mining.py	run_apriori() / run_fpgrowth()	Executes Apriori vs FP-Growth benchmarks across min_support thresholds.
src/rule_generator.py	extract_rules()	Computes Support, Conf, Lift, Leverage, Conviction. Filters Lift < 1.2 rules.
src/xai_explainer.py	explain_rule_waterfall()	Decomposes confidence into baseline P(Y) vs Lift boost using Plotly Waterfall.
app/streamlit_app.py	main() (Streamlit UI)	Dark neon UI with Admin/User auth, 3D Plotly rule topology, & cross-sell tool.

Chapter 10: Quality Assurance & Automated Pytest Suite

The system is fully validated by an automated unit test suite passing with a ****100% success rate (4/4 tests passed)**** under the supervision of Prof. Shruti Agrawal:

Test Function	Tested Pipeline Stage	Test Assertion & Verification Logic	Result Status
test_data_hygiene()	Data Loader (`src/data_loader.py`)	Asserts zero cancellation invoices ('C') and zero non-positive prices remain.	PASSED ■
test_sparse_encoder()	CSR Encoder (`src/encoder.py`)	Asserts matrix shape matches (39,517 × 5,321) and matrix is `csr_matrix` instance.	PASSED ■
test_frequent_itemsets()	Mining Engine (`src/frequent_mining.py`)	Verifies FP-Growth itemset count matches Apriori count at min_support = 0.03.	PASSED ■
test_rule_pruning()	Rule Generator (`src/rule_generator.py`)	Asserts that 100% of pruned rules satisfy Lift >= 1.2 and Confidence >= 0.5.	PASSED ■

11. Academic References & Citations

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